

1. Administrative & Study Identification

Full Study Title: An Open-Label, 2-Period, Crossover Study to Evaluate the Effects of a Single Dose of MK-1084 on the Single-Dose Pharmacokinetics of Rosuvastatin and Metformin in Healthy Participants
Short Title/Acronym: A Clinical Study of MK-1084 With Rosuvastatin and Metformin in Healthy People (MK-1084-016)
Protocol Number: MK-1084-016
NCT Number: NCT07222098
Sponsor Name: Merck
Company: Merck
Investigational Product: Calderasib (MK-1084), Rosuvastatin, Metformin
Phase of Development: Phase 1
Medical Monitor: Merck Clinical Director

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective: To evaluate the effects of a single dose of MK-1084 on the single-dose pharmacokinetics of rosuvastatin and metformin in healthy participants (comparing the amount of rosuvastatin and metformin in the body over time when given with and without MK-1084).
Secondary Objectives: To evaluate the safety and tolerability of MK-1084 when administered in combination with rosuvastatin and metformin in healthy participants, as well as additional pharmacokinetic parameters of rosuvastatin.

2.2 Background & Rationale

To understand how the investigational drug MK-1084 (Calderasib) affects the processing (absorption, breakdown, and elimination) of common metabolic medications. Rosuvastatin and metformin are well-known for managing cholesterol and blood sugar levels. This combination approach aims to evaluate potential novel mechanisms of action that could amplify metabolic profiling and safely manage conditions like metabolic syndrome.

3. Study Design & Methodology

3.1 Design Framework

Study Type: Interventional
Control Type: Active Comparator / Self-Controlled (2-Period Crossover)
Blinding: Open-label

Randomization: Crossover Assignment (Group I: Rosuvastatin + Metformin + MK-1084; Group II: Rosuvastatin + Metformin)

3.2 Planned Enrollment & Duration

Target Enrollment (\$N\$): 16 participants

Trial Status: NOT_YET_RECRUITING

Number of Sites: 1 Site (Celerion - Site 0001, Lincoln, NE, USA)

Estimated Study Start: 19-Nov-2025

Estimated Primary Completion: 14-Dec-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

- 1. Healthy individuals.
- 2. Has a Body Mass Index (BMI) ≥ 18.0 and ≤ 32.0 kg/m².
- 3. Willingness to participate in a Phase 1 clinical unit protocol involving multiple blood and urine collections.

4.2 Exclusion Criteria

- 1. Has a history of cancer (malignancy).
- 2. Positive results for human immunodeficiency virus (HIV), hepatitis B surface antigen (HBsAg), or hepatitis C virus (HCV).
- 3. Use of prohibited concomitant medications that might interfere with the absorption or metabolism of study treatments.

5. Intervention & Treatment Plan

5.1 Investigational Regimen

Dosage/Frequency: Single dose administration of MK-1084 along with single doses of Rosuvastatin and Metformin in crossover periods.

Route of Administration: Oral

Duration of Treatment: 1-day active treatment per crossover period, with an overall 4-week sample collection phase.

5.2 Concomitant & Prohibited Therapy

Permitted: Routine lifestyle medications cleared by trial coordinators.

Prohibited: Drugs known to strongly induce or inhibit relevant metabolic pathways affecting the pharmacokinetics of rosuvastatin or metformin.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	1-2 Weeks Informed Consent, Medical History, BMI evaluation, Viral Screen (HIV/HCV/HBV)
Baseline	Day 0	Randomization into Period 1 (Group I or Group II)	Interim Weeks 1-4 In-person dosing (1 day) followed

by multiple visits for intensive blood and urine sample collection to evaluate pharmacokinetics | | **End of Study** | Week 4
| Final Safety Assessment, AE evaluation, Study Discharge |

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

1. **Area Under the Concentration-Time Curve From Time 0 to Infinity (AUC_{0-inf}) of Rosuvastatin:** Blood samples will be collected at multiple time points to determine the AUC_{0-inf} of rosuvastatin.
2. **Maximum Plasma Concentration (C_{max}) of Rosuvastatin:** Blood samples will be collected at multiple time points to determine the C_{max} of rosuvastatin.
3. **AUC_{0-inf} of Metformin:** Blood samples will be collected at multiple time points to determine the AUC_{0-inf} of metformin.

7.2 Secondary Outcomes

1. **Area Under the Curve From Time 0 to Last Quantifiable Sample (AUC_{0-last}) of Rosuvastatin.**
 2. **Area Under the Curve From Time 0 to 24 Hours (AUC_{0-24hrs}) of Rosuvastatin.**
 3. **Time to Maximum Plasma Concentration (T_{max}) of Rosuvastatin.**
(Exploratory Safety Endpoints include the percentage of participants who experienced an Adverse Event (AE) or discontinued study treatment due to an AE).
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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Intensive in-clinic monitoring during the dosing and PK collection phases. AEs are collected as any unfavorable and unintended sign temporally associated with study intervention.

Serious Adverse Events (SAEs): Reported within 24 hours to the sponsor and governing IRB.

Data Monitoring Committee (DMC): Internal safety review by Merck Phase 1 clinical team.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: PK Analysis Set (for concentration metrics), Safety Analysis Set (for AE reporting).

Sample Size Justification: Target Enrollment $N = 16$ based on statistical PK powering needed for 2-period crossover drug-drug interaction (DDI) studies.

Handling of Missing Data: Standard Phase 1 PK imputation (e.g., values below the limit of quantification treated as zero for descriptive statistics).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Continuous onsite monitoring at the Phase 1 clinical facility (Celerion) to ensure precise timing of PK draws.

Audit Trail: Detailed electronic Case Report Form (eCRF) tracking of medication administration times, blood draw intervals, and AE resolution.

11. Study Identifiers & Governance

Posting ID: 12672583409764748

Primary ID: MK-1084-016

Registry Number: NCT07222098

Sponsor/Lead Organization: Merck

Study Title: A Clinical Study of MK-1084 With Rosuvastatin and Metformin in Healthy People (MK-1084-016)

Official Regulatory Title: An Open-Label, 2-Period, Crossover Study to Evaluate the Effects of a Single Dose of MK-1084 on the Single-Dose Pharmacokinetics of Rosuvastatin and Metformin in Healthy Participants

12. Clinical Framework (Drug-Drug Interaction Specific)

Primary Condition (Disease Area): Healthy (Drug-Drug Interaction Evaluation)

Diagnosis Threshold: BMI ≥ 18.0 and ≤ 32.0 kg/m²

Medical Methodology: Pharmacokinetic assessment via Crossover Design

Optimization Target: Identifying the metabolic impact of MK-1084 on lipid-lowering and glucose-lowering agents

13. Study Procedures & Methodology

Management Pathway: Intensive Phase 1 clinical unit pathway

Education Protocol (HCP): Investigator protocol training on exact PK collection intervals

Education Protocol (Patient): Onsite briefing covering meal, activity, and concurrent medication restrictions

Quality Audit Mechanism: CRO monitoring of sample processing and protocol deviation logs

14. Observation & Follow-Up Schedule

Total Duration: ~4-6 weeks (including screening and follow-up)

Onsite Visit Cadence: Multiple scheduled visits across 4 weeks

post-dose
Remote Monitoring Frequency: N/A (Intensive onsite phase 1 assessments)
Checklist Review Interval: Continuous during PK sample collection

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				